



NLP-Based Skill Gap Analyzer

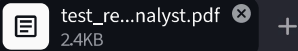
Explainable Resume and Job Description Matching powered by O*NET Occupational Data & Classical NLP.

1. Candidate Resume

Resume Input Source:

- ☒ Upload File (PDF / TXT)
- ☐ Paste Text Directly
- ☐ Load Sample Resume

Upload Resume (PDF or TXT)



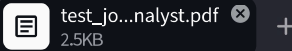
Extracted 766 characters from PDF.

2. Job Description

Job Description Input Source:

- ☒ Upload File (PDF / TXT)
- ☐ Paste Text Directly
- ☐ Load Sample Job

Upload Job Description (PDF or TXT)



Extracted 840 characters from PDF.

Analyze Match & Skill Gap

Match Summary & Gap Score

SKILL MATCH SCORE

72.7%

8 of 11 required skills

TEXT SIMILARITY

36.5%

TF-IDF Cosine Similarity

SKILL GAP LEVEL

Good Match

Candidate meets majority of requirements with minor skill gaps.

MISSING SKILLS

3

High Priority: 2

- Skill Breakdown
- Visual Analytics
- Learning Roadmap (O*NET)
- NLP Engine Inspector

Explainable NLP Pipeline Inspector

Inspect how traditional NLP algorithms processed the input step-by-step.

1. Skill Match Percentage Calculation

$$\text{Skill Match \%} = \left(\frac{|\text{Matched Skills}|}{|\text{Required Job Skills}|} \right) \times 100$$

Matched Skills Count = 8
Required Skills Count = 11
Match Percentage = (8 / 11) * 100 = 72.73%
Gap Level Threshold = Good Match

2. TF-IDF & Cosine Similarity Calculation

$$\text{Cosine Similarity}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}$$

Vector Space Model = TF-IDF (Unigrams & Bigrams, English)
Cosine Similarity (0-1) = 0.3648
Text Similarity % = 36.48%
Explanation = Calculated via TF-IDF (1-2 n-grams)

3. Preprocessed Text Preview

Lemmatized Resume Text (First 300 chars):

arjun mehta junior data analyst mumbai india arjun.mehta example.com
summary entry-level data analyst hands-on experience python sql
microsoft excel power bi panda numpy basic machine learning
interested business reporting dashboard development data-driven
decision making skill python sql microsoft ...

Lemmatized Job Description Text (First 300 chars):

job description data analyst company insightworks analytics location
mumbai india role summary looking data analyst support business
reporting data visualization analytics candidate work structured datasets
prepare dashboard communicate insight business team required skill
python sql microsoft excel...